ANSWER KEY & MARKING SCHEME SET 1

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 8531	Write the digits in decreasing order.
2	**(a)** 1358	Write the digits in increasing order (a non-zero digit first).
3	**(a)** 454	A palindrome reads the same forwards and backwards.
4	**(c)** 1000	Add 1.
5	**(c)** 999	Subtract 1.
6	**(b)** 400	Place value = digit × value of its place.
7	**(d)** even	Rule
8	**(a)** 8	Count places from the right: ones, tens, hundreds, thousands.
9	**(a)** 9642	Write the digits in decreasing order.
10	**(b)** 2469	Write the digits in increasing order (a non-zero digit first).

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Largest 9752 (1); Smallest 2579 (1).	1+1
12	Reverse of 636 is 636 (1); Yes, it is a palindrome (1).	1+1
13	Successor $= 4999 + 1 = 5000$ (1); Predecessor $= 4999 - 1 = 4998$ (1).	1+1
14	Largest 8631 (1); Smallest 1368 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Largest $= 741$, smallest $= 147$ (1); Sum $= 888$ (1); Difference $= 594$ (1).	1+1+1
16	Reverse $= 21$ (1); Sum $= 12 + 21 = 33$ (1); Yes, palindrome (1).	1+1+1
17	Middle row: $3 + 5 = 8$ and $5 + 2 = 7$ (1); Top number $= 8 + 7$ (1); $= 15$ (1).	1+1+1
	OR: Largest $= 953$, smallest $= 359$ (1); Sum $= 1312$ (1); Difference $= 594$ (1).	1+1+1
18	Reverse $= 32$ (1); Sum $= 23 + 32 = 55$ (1); Yes, palindrome (1).	1+1+1
	OR: Middle row: $4 + 1 = 5$ and $1 + 6 = 7$ (1); Top number $= 5 + 7$ (1); $= 12$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total $= 1245 + 1378 = 2623$ (1); Difference $= 1378 - 1245 = 133$ (2); Total strength is odd (2).	1+2+2
	OR: Greatest $= 8642$ (1); Smallest $= 2468$ (1); Difference $= 6174$ (1); Even (2).	1+1+1+2
20	Total $= 2310 + 1795 = 4105$ (1); Difference $= 2310 - 1795 = 515$ (2); Total strength is odd (2).	1+2+2
	OR: Greatest $= 7531$ (1); Smallest $= 1357$ (1); Difference $= 6174$ (1); Even (2).	1+1+1+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 3223	A palindrome reads the same forwards and backwards.
2	**(c)** 4570	Add 1.
3	**(d)** 5229	Subtract 1.
4	**(c)** 70000	Place value = digit × value of its place.
5	**(a)** even	Rule
6	**(a)** 9	Count places from the right: ones, tens, hundreds, thousands.
7	**(b)** 7530	Write the digits in decreasing order.
8	**(b)** 3057	Write the digits in increasing order (a non-zero digit first).
9	**(a)** 7887	A palindrome reads the same forwards and backwards.
10	**(d)** 10100	Add 1.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Reverse of 1234 is 4321 (1); No, it is not a palindrome (1).	1+1
12	Successor = $10000 + 1 = 10001$ (1); Predecessor = $10000 - 1 = 9999$ (1).	1+1
13	Largest 9420 (1); Smallest 2049 (1).	1+1
14	Reverse of 4994 is 4994 (1); Yes, it is a palindrome (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Largest = 862, smallest = 268 (1); Sum = 1130 (1); Difference = 594 (1).	1+1+1
16	Reverse = 43 (1); Sum = $34 + 43 = 77$ (1); Yes, palindrome (1).	1+1+1
17	Middle row: $7 + 2 = 9$ and $2 + 5 = 7$ (1); Top number = $9 + 7$ (1); = 16 (1).	1+1+1
	OR: Largest = 731, smallest = 137 (1); Sum = 868 (1); Difference = 594 (1).	1+1+1
18	Reverse = 14 (1); Sum = $41 + 14 = 55$ (1); Yes, palindrome (1).	1+1+1
	OR: Middle row: $6 + 3 = 9$ and $3 + 8 = 11$ (1); Top number = $9 + 11$ (1); = 20 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total = $1650 + 2284 = 3934$ (1); Difference = $2284 - 1650 = 634$ (2); Total strength is even (2).	1+2+2
	OR: Greatest = 9721 (1); Smallest = 1279 (1); Difference = 8442 (1); Even (2).	1+1+1+2
20	Total = $3127 + 1856 = 4983$ (1); Difference = $3127 - 1856 = 1271$ (2); Total strength is odd (2).	1+2+2
	OR: Greatest = 8630 (1); Smallest = 3068 (1); Difference = 5562 (1); Even (2).	1+1+1+2

Mistakes to avoid:

- Writing 0 first when forming the smallest number (e.g. 046); the smallest 3-digit number from 4, 0, 6 is 406.
- In a Kaprekar step, subtracting the larger from the smaller or adding the two arrangements.
- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
- Reversing digits wrongly in reverse-and-add (the reverse of 68 is 86) or stopping one step early.
- Counting the starting number as a step when counting steps to reach 1 or a palindrome.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 3: **Number Play** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 24000	Subtract 1.
2	**(b)** 30	Place value = digit × value of its place.
3	**(a)** odd	Rule
4	**(c)** 0	Count places from the right: ones, tens, hundreds, thousands.
5	**(a)** 8641	Write the digits in decreasing order.
6	**(c)** 1468	Write the digits in increasing order (a non-zero digit first).
7	**(c)** 12521	A palindrome reads the same forwards and backwards.
8	**(c)** 79000	Add 1.
9	**(d)** 89999	Subtract 1.
10	**(b)** 5000	Place value = digit × value of its place.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Successor $= 36070 + 1 = 36071$ (1); Predecessor $= 36070 - 1 = 36069$ (1).	1+1
12	Largest 7520 (1); Smallest 2057 (1).	1+1
13	Reverse of 5605 is 5065 (1); No, it is not a palindrome (1).	1+1
14	Successor $= 89999 + 1 = 90000$ (1); Predecessor $= 89999 - 1 = 89998$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Largest $= 854$, smallest $= 458$ (1); Sum $= 1312$ (1); Difference $= 396$ (1).	1+1+1
16	Reverse $= 25$ (1); Sum $= 52 + 25 = 77$ (1); Yes, palindrome (1).	1+1+1
17	Middle row: $2 + 9 = 11$ and $9 + 4 = 13$ (1); Top number $= 11 + 13$ (1); $= 24$ (1).	1+1+1
	OR: Largest $= 962$, smallest $= 269$ (1); Sum $= 1231$ (1); Difference $= 693$ (1).	1+1+1
18	Reverse $= 16$ (1); Sum $= 61 + 16 = 77$ (1); Yes, palindrome (1).	1+1+1
	OR: Middle row: $8 + 5 = 13$ and $5 + 7 = 12$ (1); Top number $= 13 + 12$ (1); $= 25$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Total $= 2094 + 2511 = 4605$ (1); Difference $= 2511 - 2094 = 417$ (2); Total strength is odd (2).	1+2+2
	OR: Greatest $= 9541$ (1); Smallest $= 1459$ (1); Difference $= 8082$ (1); Even (2).	1+1+1+2
20	Total $= 1788 + 1949 = 3737$ (1); Difference $= 1949 - 1788 = 161$ (2); Total strength is odd (2).	1+2+2
	OR: Greatest $= 8732$ (1); Smallest $= 2378$ (1); Difference $= 6354$ (1); Even (2).	1+1+1+2

Mistakes to avoid:

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- Applying the odd rule ($3n + 1$) to an even number or halving an odd number in the hailstone game.
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